SEAA 4243 FINITE ELEMENT METHOD · 2L + 1T

2D Heat Transfer Part 2 of 2

From the weak form to the solution: [k], assembly, Dirichlet conditions, post-processing and the MATLAB code



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Text: Mohd Yassin et al., *Two-Dimensional Finite Element Formulations for Heat, Solids and Fluids (with Matlab)*, MSNM 2020 — §2.2.2 – §2.4

ACTIVITY 0 · INDIVIDUAL · RETRIEVAL WARM-UP

Three questions from the last session — answer on paper before we start

5 MIN

Write down

- The steady heat conduction PDE in matrix form, and the two matrices it contains.
- The name of the boundary condition that needs no action in FEM, and why.
- N_2 of the 4-node element and its value at node 3.

$$T = N_1\hat{T}_1 + N_2\hat{T}_2 + N_3\hat{T}_3 + N_4\hat{T}_4 = \{N\}\{\hat{T}\}^T$$

WHY No notes. Retrieval practice is the point: what you can recall is what you can use in the exam.

5 minutes. Then we compare with the slides.



LEARNING OUTCOMES · TODAY

By the end of this session you should be able to

1. Apply the Galerkin weighted residual method and integration by parts to obtain $[k]\{\hat{T}\}^T = \{q\} + \{b\}$ and explain what each term means.
2. Evaluate the conductance matrix [k] and the load vectors {q} and {b} of a 4-node rectangular element in closed form.
3. Assemble element matrices into the global system with a connectivity table and impose Dirichlet conditions by row–column elimination.
4. Solve a small problem by hand, post-process temperature and flux at a point, and judge the result against a refined mesh.

CHECK All four outcomes are examined by Assignment 4 (released today) and by the final exam under CLO3.



SESSION MAP · 180 MINUTES

Six blocks, seven short activities, one break

0:00	0:30	1:05	1:25	1:35	2:15	2:40
Weak form & IBP	**Element matrices**	**Mesh & connectivity**	**Break**	**Assembly & Dirichlet**	**Post-processing**	**MATLAB & Asg 4**
30 min	35 min	20 min	10 min	40 min	25 min	20 min
A1 · 6 min	**A2 · 8 min**	**A3 · 6 min**	no activity	**A4 + A5**	**A6 + QR**	**Exit ticket**

Bring forward from the last session

- Your Activity 4–5 sheet: the same element, the same numbers.
- The formula sheet: $N_1 \ldots N_4$ and their derivatives.
- A laptop with MATLAB for the code trace in Block 6.

Today's worked example

- Text §2.3: 1 m × 1 m plate, $q_h = 600$ W m^{-3}, k = 300 W m^{-1} K^{-1}.
- Four 4-node elements by hand (2 × 2 mesh).
- Then 16 × 16 in MATLAB and a comparison with software.



BLOCK 1 · 0:00 – 0:30

1

From residual to weak form

Substitute $T = \{N\}\{\hat{T}\}^T$ into the PDE, weight the residual with $\{N\}^T$, integrate by parts. Text §2.2.2.



BLOCK 1 · SUBSTITUTE THE APPROXIMATION · TEXT (2.34)

Substituting the approximation leaves a residual

$$\{\partial\}[E]\{\partial\}^T\{N\}\{\hat{T}\}^T \neq -q_h \quad (2.34)$$

Count

Four unknowns $\hat{T}_1 \ldots \hat{T}_4$, but only one equation — and it holds nowhere exactly.

Idea

Make the residual zero *on average*, weighted by each shape function: four weights, four equations.

$$\int_x \int_y \{N\}^T \left(\{\partial\}[E]\{\partial\}^T\{N\}\{\hat{T}\}^T + q_h \right) dy\, dx = 0 \quad (2.35)$$ Galerkin: weights = shape functions

WHY Galerkin: the weights are the shape functions themselves, which makes [k] symmetric.



BLOCK 1 · INTEGRATION BY PARTS: THE 1D CASE YOU KNOW

Move one derivative from T onto the weight; a boundary term appears

$$\int_0^L N_i \frac{d}{dx}\left(k\frac{dT}{dx}\right)dx = \left[N_i\, k\frac{dT}{dx}\right]_0^L - \int_0^L \frac{dN_i}{dx}\, k\, \frac{dT}{dx}\, dx$$

bar element, earlier in the course

Purpose 1 · relax continuity

After IBP only first derivatives of N and T appear, so a piecewise-linear (or bilinear) T is admissible. Second derivatives of a bilinear field would be zero or undefined at element edges.

Purpose 2 · expose the natural condition

The boundary term contains k dT/dx, the flux. Where the flux is specified it becomes a load; where it is zero it vanishes; where T is specified the term is removed with the equation.



Green’s theorem turns the area integral of a derivative into a boundary integral

$$\int_x\int_y N_i \frac{\partial F}{\partial x}\, dy\, dx = \oint_s N_i\, F\, n_x\, ds - \int_x\int_y \frac{\partial N_i}{\partial x}\, F\, dy\, dx$$

Green’s theorem, x-direction (same for y)

$$-\int_x\int_y \{N\}^T \{\partial\} [E] \{\partial\}^T \{N\} \{\hat{T}\}^T\, dy\, dx = -\int_x\int_y \{N\}^T q_h\, dy\, dx - \int_s \{N\}^T q_b\, ds \quad (2.36)$$

WHY n_x, n_y are the outward normal components and ds runs around the element boundary. Apply it to each row of $\{\partial\}[E]\{\partial\}^T$.

WATCH OUT $q_b = k\, \partial T/\partial n$ is the boundary flux of (2.15), positive when heat **enters**. Text (2.41) writes the components with the opposite sign — use (2.15).



BLOCK 1 · REARRANGE AND NAME THE TERMS · TEXT (2.37)–(2.42)

Conductance matrix, heat-generation vector, boundary-flux vector

$$\underbrace{\int_x\int_y \{N\}^T\{\partial\}[E]\{\partial\}^T\{N\}\,dy\,dx}_{[k]}\{\hat{T}\}^T = \underbrace{\int_x\int_y \{N\}^T q_h\,dy\,dx}_{\{q\}} + \underbrace{\int_s \{N\}^T q_b\,ds}_{\{b\}} \quad (2.37)$$

$$q_b = \left(k_x\frac{\partial T}{\partial x}\right)n_x + \left(k_y\frac{\partial T}{\partial y}\right)n_y = k\,\frac{\partial T}{\partial n} \qquad \text{(positive when heat enters)} \quad (2.15)$$

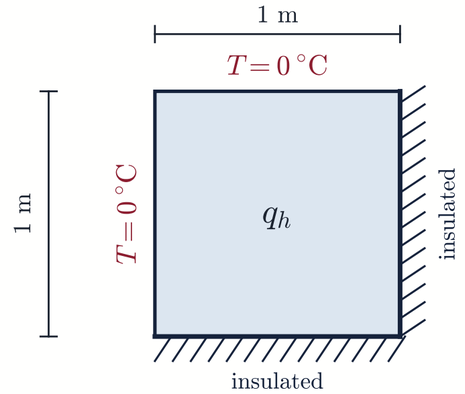
$$[k]\{\hat{T}\}^T = \{q\} + \{b\} = \{r\} \quad (2.42)$$

WHY Read [k]'s integrand left to right: {N}ᵀ{∂} means the derivative acts on the weight {N}ᵀ (moved there by the IBP). {∂}ᵀ{N} is the last session's 2 × 4 gradient matrix.

ACTIVITY 1 · PAIRS · REASONING

Which term, which edge? Trace the boundary term around the plate

6 MIN



For the text's plate (top and left at 0 °C, right and bottom insulated)

- Which edges contribute to {b}? Write the value of q_b on each.
- On the two Dirichlet edges the flux is unknown — what happens to the boundary term there?
- How many equations does the weighted residual give for one 4-node element, and for the 2 × 2 mesh before boundary conditions?

6 minutes. Reveal follows.

BLOCK 2 · 0:30 – 1:05

2 The element matrices

Integrate in closed form over the rectangle: k_{34} step by step, then the whole [k], then {q} and {b}. Text (2.43)–(2.50).



BLOCK 2 · K_{34} IN FOUR SYMBOLIC STEPS · TEXT (2.43)–(2.46)

Row 3 of {N}ᵀ, the operator, [E], the operator, column 4 of {N}

$$\cdots \{N_3\} \begin{Bmatrix} \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} \end{Bmatrix} \cdots$$ Step 1

$$\cdots \begin{Bmatrix} \dfrac{\partial N_3}{\partial x} & \dfrac{\partial N_3}{\partial y} \end{Bmatrix} \begin{bmatrix} k_x & 0 \\ 0 & k_y \end{bmatrix} \cdots$$ Step 2

$$\cdots \begin{Bmatrix} k_x \dfrac{\partial N_3}{\partial x} & k_y \dfrac{\partial N_3}{\partial y} \end{Bmatrix} \begin{Bmatrix} \dfrac{\partial}{\partial x} \\ \dfrac{\partial}{\partial y} \end{Bmatrix} \cdots$$ Step 3

$$\cdots \begin{Bmatrix} k_x \dfrac{\partial N_3}{\partial x} \dfrac{\partial}{\partial x} & k_y \dfrac{\partial N_3}{\partial y} \dfrac{\partial}{\partial y} \end{Bmatrix} \{N_4\} \cdots$$ Step 4

What happened

- Step 1 → 2: the operator acts on N_3 to its left (IBP moved it there).
- Step 2 → 3: multiply by [E]; k_x and k_y attach to the two derivatives.
- Step 3 → 4: the second operator acts to the right, on N_4.
- Step 5: integrate over the rectangle (next slide).



Two separable integrals over 0 ≤ x ≤ a, 0 ≤ y ≤ b

$$k_{34} = \int_0^a \int_0^b \left(k_x \frac{\partial N_3}{\partial x} \frac{\partial N_4}{\partial x} + k_y \frac{\partial N_3}{\partial y} \frac{\partial N_4}{\partial y} \right) dy\, dx \tag{2.47}$$

$$\frac{\partial N_3}{\partial x} \frac{\partial N_4}{\partial x} = \frac{y}{ab} \cdot \frac{-y}{ab} = -\frac{y^2}{a^2 b^2} \qquad \Rightarrow \int_0^a \int_0^b -\frac{k_x\, y^2}{a^2 b^2}\, dy\, dx = -\frac{b\, k_x}{3a}$$

$$\frac{\partial N_3}{\partial y} \frac{\partial N_4}{\partial y} = \frac{x}{ab} \cdot \frac{a - x}{ab} = \frac{x(a - x)}{a^2 b^2} \qquad \Rightarrow \int_0^a \int_0^b \frac{k_y\, x(a - x)}{a^2 b^2}\, dy\, dx = \frac{a\, k_y}{6b}$$

$$k_{34} = -\frac{b\, k_x}{3a} + \frac{a\, k_y}{6b} \tag{2.48}$$



Four distinct values; symmetric; every row sums to zero

$$[k] = \begin{bmatrix} s_1 & s_2 & s_3 & s_4 \\ s_2 & s_1 & s_4 & s_3 \\ s_3 & s_4 & s_1 & s_2 \\ s_4 & s_3 & s_2 & s_1 \end{bmatrix} \tag{2.49}$$

$$s_1 = \frac{b\,k_x}{3a} + \frac{a\,k_y}{3b}, \qquad s_2 = -\frac{b\,k_x}{3a} + \frac{a\,k_y}{6b}$$

$$s_3 = -\frac{b\,k_x}{6a} - \frac{a\,k_y}{6b}, \qquad s_4 = \frac{b\,k_x}{6a} - \frac{a\,k_y}{3b}$$

$$s_1 + s_2 + s_3 + s_4 = 0 \quad \Rightarrow \quad [k]\begin{Bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

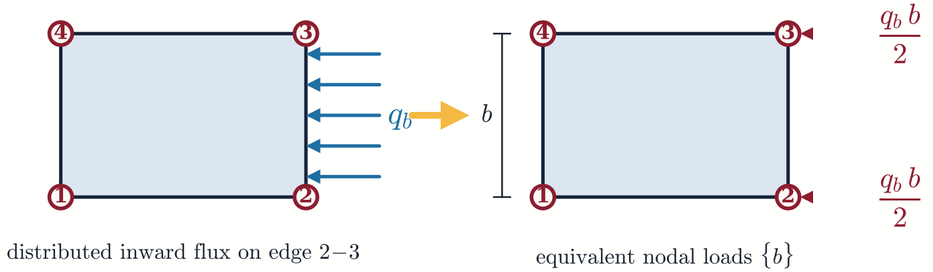
CHECK Row sums are zero because a uniform temperature produces no flux: [k]{1} = {0}. Use it as a check on every hand calculation.

WATCH OUT [k] is singular on its own (one zero eigenvalue). Only the Dirichlet conditions make the system solvable.

Uniform generation splits equally; a uniform edge flux splits between the two edge nodes

$$\{q\} = q_h \int_0^a \int_0^b \{N\}^T dy\, dx = \frac{q_h\, a\, b}{4} \begin{Bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{Bmatrix} \quad (2.39)$$

$$\{b\} = \int_0^b \{N\}^T \big|_{x=a}\, q_b\, dy = q_b \int_0^b \begin{Bmatrix} 0 & \frac{b-y}{b} & \frac{y}{b} & 0 \end{Bmatrix}^T dy = \frac{q_b\, b}{2} \begin{Bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{Bmatrix} \quad (2.40)$$



distributed inward flux on edge 2–3 — equivalent nodal loads $\{b\}$

WHY {b} is assembled only on the elements whose edge lies on the flux boundary; interior edges never contribute.

CHECK Dimensions with unit thickness: $q_h ab/4$ is W m⁻³ × m² = W per metre of thickness; $q_b b/2$ likewise.

ACTIVITY 2 · PAIRS · CALCULATION

Evaluate [k] and {q} for the text's element, then check the row sums

8 MIN

Given (text §2.3)

- a = b = 0.5 m (2 × 2 mesh of a 1 m × 1 m plate), unit thickness.
- $k_x = k_y = 300$ W m^{-1} K^{-1}, $q_h = 600$ W m^{-3}.

Find

- s_1, s_2, s_3, s_4 and hence [k].
- {q} for the element.
- Verify that each row of [k] sums to zero.

$$s_1 = \frac{b\,k_x}{3a} + \frac{a\,k_y}{3b}, \qquad s_2 = -\frac{b\,k_x}{3a} + \frac{a\,k_y}{6b}$$

$$s_3 = -\frac{b\,k_x}{6a} - \frac{a\,k_y}{6b}, \qquad s_4 = \frac{b\,k_x}{6a} - \frac{a\,k_y}{3b}$$

CHECK Units: W K^{-1} for [k]; W for {q} (per metre of thickness).

8 minutes. Early finishers: what changes in [k] if a is doubled and b kept?



ACTIVITY 2 · SOLUTION · TEXT (2.51)–(2.52)

The same [k] and {r} for all four elements of the mesh

$$s_1 = \frac{0.5(300)}{3(0.5)} + \frac{0.5(300)}{3(0.5)} = 100 + 100 = 200, \qquad s_2 = -100 + 50 = -50$$

$$s_3 = -50 - 50 = -100, \qquad s_4 = 50 - 100 = -50$$

$$[k^1] = [k^2] = [k^3] = [k^4] = \begin{bmatrix} 200 & -50 & -100 & -50 \\ -50 & 200 & -50 & -100 \\ -100 & -50 & 200 & -50 \\ -50 & -100 & -50 & 200 \end{bmatrix} \quad (2.51)$$

$$\{r^1\} = \{r^2\} = \{r^3\} = \{r^4\} = \frac{600\,(0.5)(0.5)}{4} \begin{Bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 37.5 \\ 37.5 \\ 37.5 \\ 37.5 \end{Bmatrix}$$

Right: (2.52), the heat-generation vector of every element.

BLOCK 3 · 1:05 – 1:25

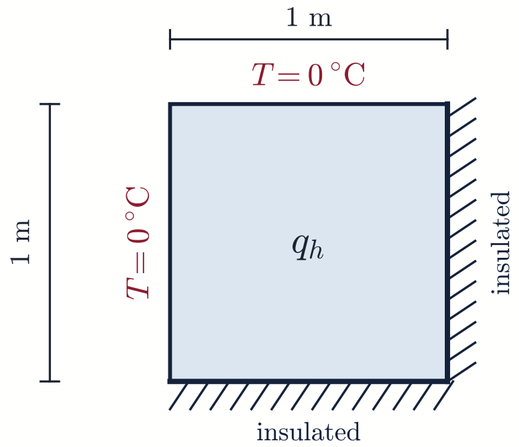
3

Mesh and connectivity

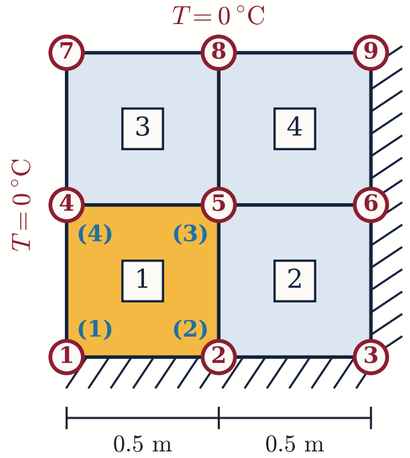
Number the global nodes, list each element’s nodes in local order, and you have the map from local to global. Text Fig. 2.6.



A 1 m square plate meshed with four elements



q_h = 600 W m⁻³, k = 300 W m⁻¹ K⁻¹, unit thickness.



Four elements, nine global nodes. Element 1 shaded; its local nodes (1)…(4) sit at global nodes 1, 2, 5, 4.

WHY Heat source, two cold edges, two insulated edges. Coarse on purpose: the hand calculation must be traceable; Block 5 refines it.

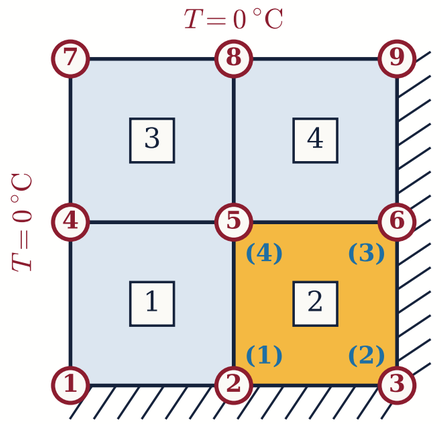


BLOCK 3 · THE CONNECTIVITY TABLE

Element e, local node i → global node conn(e, i): always anticlockwise from the bottom-left

Element	(1)	(2)	(3)	(4)
1	1	2	5	4
2	2	3	6	5
3	4	5	8	7
4	5	6	9	8

WHY The MATLAB line K([1 2 5 4],[1 2 5 4]) = K([1 2 5 4],[1 2 5 4]) + k is this table's first row.

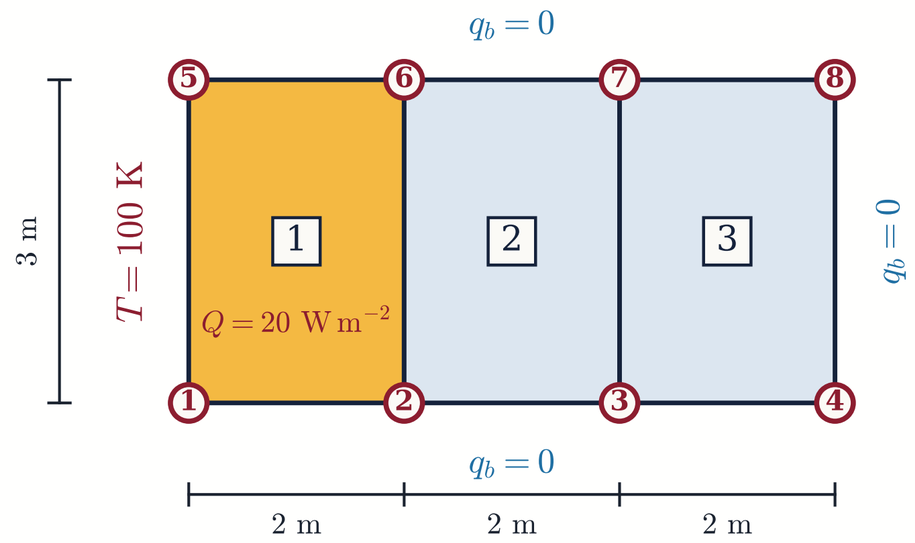


Element 2: local (1)…(4) = global 2, 3, 6, 5. Node 5 is local 4 here, local 3 in element 1: the same unknown T_5.

ACTIVITY 3 · PAIRS · BOOKKEEPING

Write the connectivity table for the Assignment 4 strip

6 MIN



Do

- List the global nodes of elements 1, 2 and 3 in local order (1)…(4).
- Mark which global nodes have a known temperature.
- How many equations remain after imposing them?

Keep this table — you will need it for Assignment 4.

6 minutes, then a 10-minute break.

ACTIVITY 3 · SOLUTION

Three rows, two known nodes, six unknowns

Element	(1)	(2)	(3)	(4)
1	1	2	6	5
2	2	3	7	6
3	3	4	8	7

Known and unknown

- Known: $T_1 = T_5 = 100$ K (left edge).
- Unknown: T_2, T_3, T_4, T_6, T_7, T_8 — six equations.
- Insulated edges: nothing to do. Heat source on element 1 only: $\{r^1\} \neq 0$, $\{r^2\} = \{r^3\} = 0$.



1:25 – 1:35

Break

Back in 10 minutes. Second half: assembly, Dirichlet conditions, solution, post-processing and MATLAB.

BLOCK 4 · 1:35 – 2:15

4

Assembly and Dirichlet conditions

Scatter each [k^e] into [K] with the connectivity table, then remove the known temperatures. Text (2.53)–(2.55).



BLOCK 4 · THE SCATTER-ADD RULE

Each local entry is added to the global position that the connectivity table names

$$K_{IJ} \;+\!=\; k^{e}_{ij} \qquad \text{with } I = \mathrm{conn}(e,i),\; J = \mathrm{conn}(e,j)$$

In words

- Start with [K] = 0 (9 × 9) and {R} = 0 (9 × 1).
- For each element: read its row of the table; add k_{ij} to K_{IJ} and r_i to R_I.
- Shared nodes receive contributions from every element that touches them.

Why ‘add’

Each element’s weighted-residual equations are statements about its own area; the equation for a global node sums the statements of all elements around it — heat balance at that node.

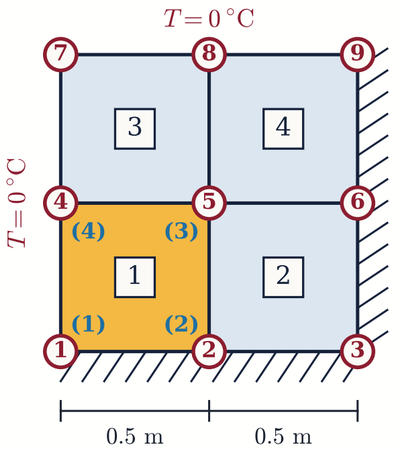
CHECK Assembled [K] is symmetric, sparse (zeros where nodes are not connected) and still singular until Dirichlet conditions are imposed.



BLOCK 4 · AFTER ELEMENT 1 · NODES [1 2 5 4]

Sixteen entries land in rows and columns 1, 2, 4, 5

$$\begin{bmatrix} 200 & -50 & 0 & -50 & -100 & 0 & 0 & 0 & 0 \\ -50 & 200 & 0 & -100 & -50 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -50 & -100 & 0 & 200 & -50 & 0 & 0 & 0 & 0 \\ -100 & -50 & 0 & -50 & 200 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



Maroon: element 1. Local (1)…(4) → global 1, 2, 5, 4: local $k_{23} = s_4 = -50$ goes to K_{25}; local $k_{13} = -100$ goes to K_{15}.

ACTIVITY 4 · PAIRS · ASSEMBLY

Add element 2 (nodes [2 3 6 5]) to [K]

8 MIN

$$\begin{bmatrix} 200 & -50 & 0 & -50 & -100 & 0 & 0 & 0 & 0 \\ -50 & 200 & 0 & -100 & -50 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -50 & -100 & 0 & 200 & -50 & 0 & 0 & 0 & 0 \\ -100 & -50 & 0 & -50 & 200 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

[K] after element 1 (maroon).

Do

- Write the 10 distinct entries that element 2 (nodes 2, 3, 6, 5) adds; its [k] equals element 1’s.
- Which four entries now have contributions from both elements? Give their totals.
- Check: row 2 of [K] must still sum to zero.

8 minutes. Reveal follows.



ACTIVITY 4 · SOLUTION · AFTER ELEMENTS 1 AND 2

Shared entries are sums; the rest come from one element

$$\begin{bmatrix} 200 & -50 & 0 & -50 & -100 & 0 & 0 & 0 & 0 \\ -50 & 400 & -50 & -100 & -100 & -100 & 0 & 0 & 0 \\ 0 & -50 & 200 & 0 & -100 & -50 & 0 & 0 & 0 \\ -50 & -100 & 0 & 200 & -50 & 0 & 0 & 0 & 0 \\ -100 & -100 & -100 & -50 & 400 & -50 & 0 & 0 & 0 \\ 0 & -100 & -50 & 0 & -50 & 200 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Maroon: element 1 · teal: element 2 · green: both elements add.

Totals at the shared nodes

- $K_{22} = 200 + 200 = 400$
- $K_{55} = 200 + 200 = 400$
- $K_{25} = K_{52} = -50 - 50 = -100$

CHECK Row 2: $-50 + 400 - 50 - 100 - 100 - 100 = 0$.

Nine equations, nine unknowns — but five temperatures are already known

$$\begin{bmatrix} 200 & -50 & 0 & -50 & -100 & 0 & 0 & 0 & 0 \\ -50 & 400 & -50 & -100 & -100 & -100 & 0 & 0 & 0 \\ 0 & -50 & 200 & 0 & -100 & -50 & 0 & 0 & 0 \\ -50 & -100 & 0 & 400 & -100 & 0 & -50 & -100 & 0 \\ -100 & -100 & -100 & -100 & 800 & -100 & -100 & -100 & -100 \\ 0 & -100 & -50 & 0 & -100 & 400 & 0 & -100 & -50 \\ 0 & 0 & 0 & -50 & -100 & 0 & 200 & -50 & 0 \\ 0 & 0 & 0 & -100 & -100 & -100 & -50 & 400 & -50 \\ 0 & 0 & 0 & 0 & -100 & -50 & 0 & -50 & 200 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \\ T_7 \\ T_8 \\ T_9 \end{Bmatrix} = \begin{Bmatrix} 37.5 \\ 75 \\ 37.5 \\ 75 \\ 150 \\ 75 \\ 37.5 \\ 75 \\ 37.5 \end{Bmatrix} \quad (2.53)$$

Load vector: 37.5 at corner nodes (one element), 75 at edge nodes (two elements), 150 at the centre node (four elements).



BLOCK 4 · IMPOSE THE KNOWN TEMPERATURES

Known T: delete the equation, move the value right

Row–column elimination ($\bar{T} = 0$)

- Known: $T_1 = T_4 = T_7 = T_8 = T_9 = 0$ (top and left edges).
- Delete rows and columns 1, 4, 7, 8, 9 of [K] and rows of {R}.
- Keep rows and columns 2, 3, 5, 6: a 4 × 4 system.

Non-zero $\bar{T}$ (e.g. 100 K in Assignment 4)

$$\{F\} \leftarrow \{F\} - [K]_{(:,\,k)}\,\bar{T}_k \qquad \text{then delete row and column } k$$

The known column times $\bar{T}$ is a known heat flow; it moves to the right-hand side before the row and column are deleted.
MATLAB: F = F – K(:,dof_k)*To.

$$\begin{bmatrix} 400 & -50 & -100 & -100 \\ -50 & 200 & -100 & -50 \\ -100 & -100 & 800 & -100 \\ -100 & -50 & -100 & 400 \end{bmatrix} \begin{Bmatrix} T_2 \\ T_3 \\ T_5 \\ T_6 \end{Bmatrix} = \begin{Bmatrix} 75 \\ 37.5 \\ 150 \\ 75 \end{Bmatrix} \quad \Rightarrow \quad \begin{Bmatrix} T_2 \\ T_3 \\ T_5 \\ T_6 \end{Bmatrix} = \begin{Bmatrix} 0.482 \\ 0.621 \\ 0.386 \\ 0.482 \end{Bmatrix}$$

(2.54) → (2.55)

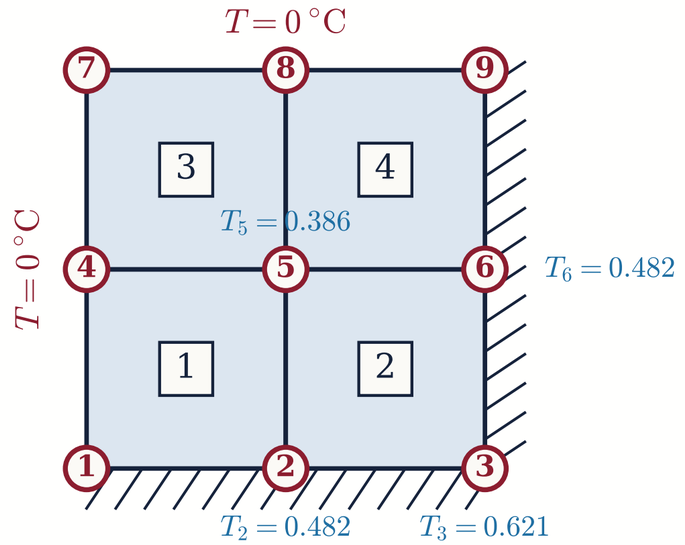
ACTIVITY 5 · PAIRS · VERIFICATION

Verify the solution and read it physically

6 MIN

Do

- Substitute $T_2 = 0.482$, $T_3 = 0.621$, $T_5 = 0.386$, $T_6 = 0.482$ into row 1 of the reduced system (2.54). Does it give 75?
- Why is T_3 the largest and T_5 the smallest of the four?
- Why does T_2 equal T_6 exactly?



6 minutes. Reveal follows.

ACTIVITY 5 · SOLUTION

Row 1 closes to within rounding; the pattern of the answer matches the physics

$$400(0.482) - 50(0.621) - 100(0.386) - 100(0.482) = 192.8 - 31.05 - 38.6 - 48.2$$
$$= 74.95 \approx 75 \checkmark$$

Hottest at node 3

Farthest from the two cold edges, enclosed by the two insulated edges: the prediction from Heat Transfer Part 1 is confirmed.

$T_2 = T_6$

Symmetry about the diagonal through nodes 1, 5, 9: swapping x and y maps the problem onto itself, so the solution must be symmetric too.

Use symmetry in exams

If a problem is symmetric, equal temperatures are equal unknowns: fewer equations to solve by hand.



BLOCK 5 · 2:15 – 2:40

5

Post-processing and validation

Temperature and flux anywhere in the plate, then the honest question: how wrong is a 2 × 2 mesh? Text Table 2.1, Fig. 2.8.



Find the element, go local, interpolate with its own nodal values

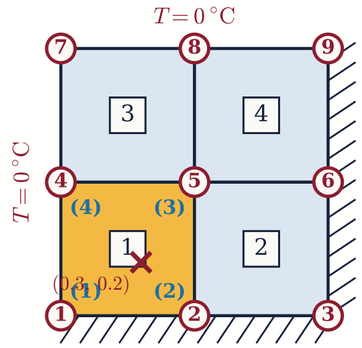
1. Locate the element that contains the point (from the mesh geometry).
2. Local coordinates: subtract the coordinates of that element's node 1.

$$x_{\text{loc}} = x - x_1^e, \qquad y_{\text{loc}} = y - y_1^e$$

3. Interpolate with the shape functions and the element's four nodal temperatures, in local order.

$$T(x,y) = \{N(x_{\text{loc}}, y_{\text{loc}})\}\{\hat{T}^e\}^T$$

$$\{\hat{T}^e\} = \{T_{\text{conn}(e,1)} \;\; T_{\text{conn}(e,2)} \;\; T_{\text{conn}(e,3)} \;\; T_{\text{conn}(e,4)}\}$$



WATCH OUT Nodal values in the element's local order (1)…(4) = its connectivity row, not global order.

BLOCK 5 · FLUX AT A POINT

Differentiate the same interpolation; the flux is constant along one direction in each element

$$\begin{Bmatrix} q_x \\ q_y \end{Bmatrix} = -[E]\{\partial\}^T\{N\}\{\hat{T}^e\}^T \qquad \text{element 2, centroid: } \begin{Bmatrix} q_x \\ q_y \end{Bmatrix} = \begin{Bmatrix} -70.7 \\ +70.7 \end{Bmatrix} \mathrm{W\,m^{-2}}$$

Reading the sign

At the centroid of element 2 (nodes 2, 3, 6, 5): $q_x < 0$, heat flows towards –x, from the hot corner at node 3 towards node 2; $q_y > 0$, towards the cold top edge.

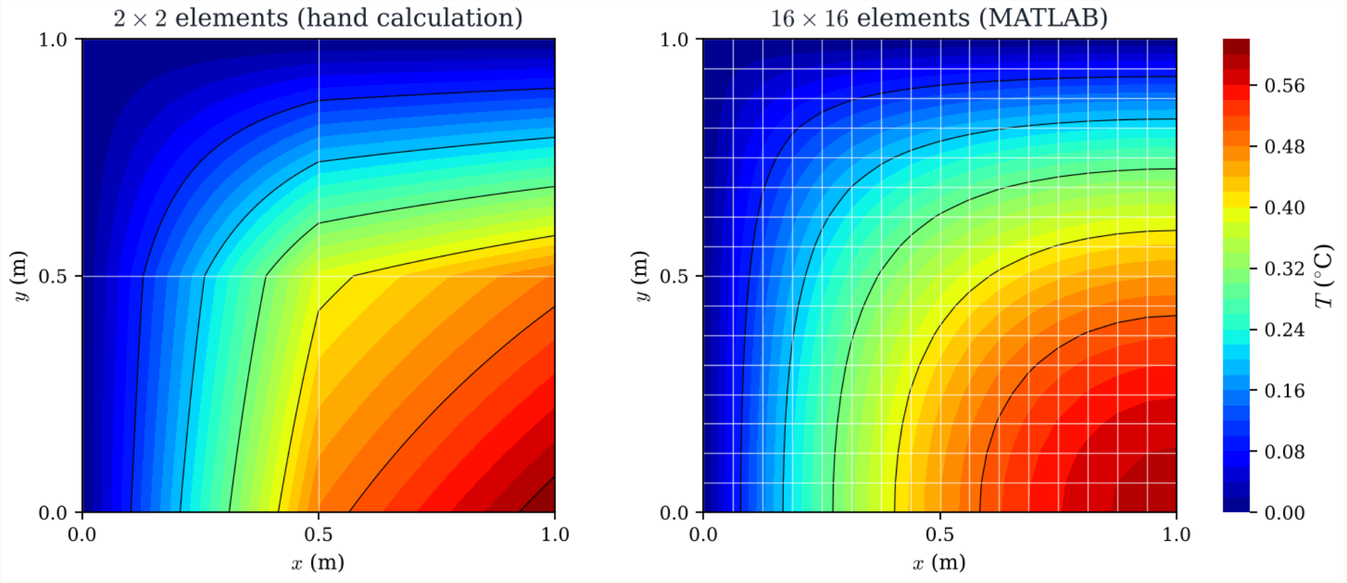
Discontinuity

The flux jumps across element edges because the gradient of a bilinear field is not continuous. Software averages nodal fluxes; the size of the jump is a free error indicator.



BLOCK 5 · HOW WRONG IS A 2 × 2 MESH? · TEXT TABLE 2.1, FIG. 2.8

The 2 × 2 mesh is about 5% high; 16 × 16 matches software



Mesh	T(L/2, 0)	T(L, 0)
Q4 2 × 2 (hand)	0.4821	0.6214
Q8 2 × 1 (hand)	0.4448	0.5836
Q4 16 × 16	0.4590	0.5898
Software (text)	0.4589	0.5897

CHECK Convergence check = the same problem on a finer mesh; refine until the answer stops moving. the ANSYS demo's ANSYS exercise.

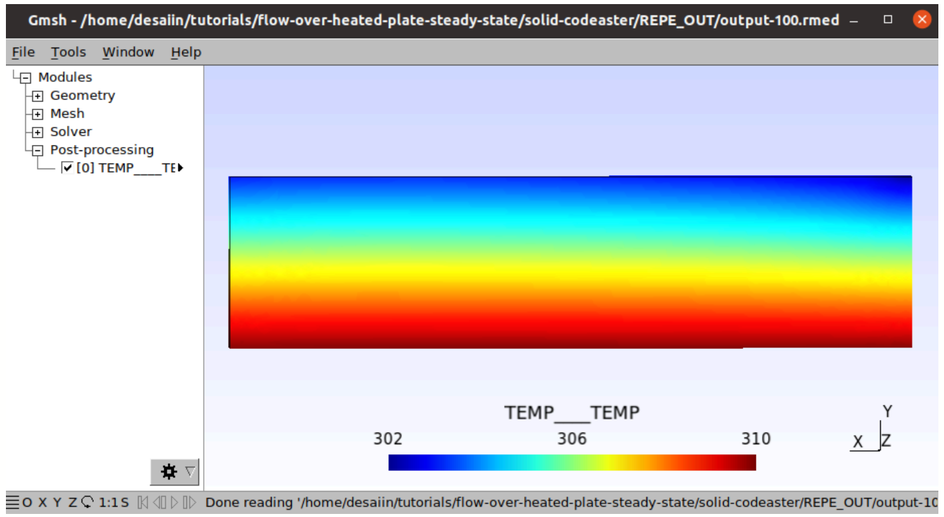
$$\text{error} = \frac{T_{2\times2} - T_{\text{ref}}}{T_{\text{ref}}}$$

$$\frac{0.4821 - 0.4589}{0.4589} = +5.1\%, \qquad \frac{0.6214 - 0.5897}{0.5897} = +5.4\%$$

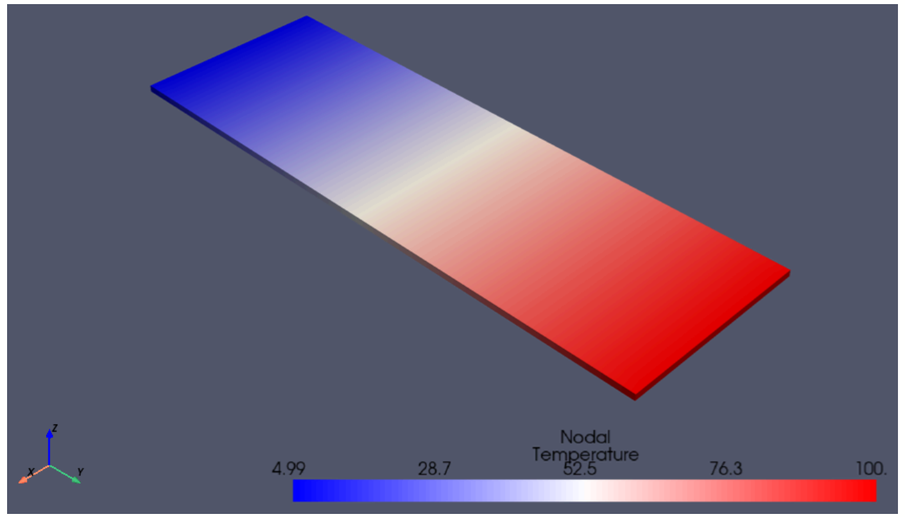


BLOCK 5 · THE SAME STEADY CONDUCTION IN OTHER FE CODES

Same equations, same checks, other codes



Code_Aster (open-source FE), solid plate heated from below, shown in Gmsh. preCICE tutorials, “Flow over a heated plate, steady state”, LGPL-3.0.



ANSYS MAPDL, plate with two faces at 5 and 100 °C. PyMAPDL documentation, “Basic thermal analysis”, © ANSYS, Inc., MIT licence.

WHY Left: the field your Heat Transfer Part 2 script would produce for that plate — hot boundary below, cooled boundary above, contours parallel to the edges.

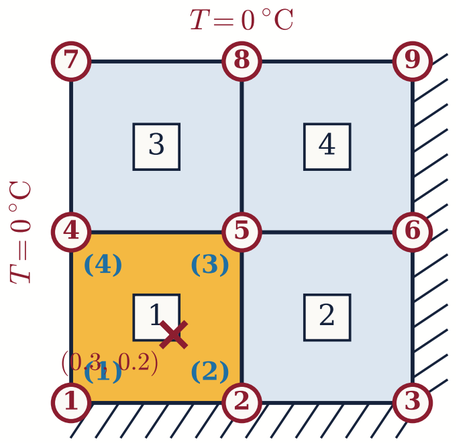
CHECK Right: no source and two fixed faces give a linear T. Every FE code must reproduce it exactly, because the four-node element contains 1, x and y: a quick check of any new model.



ACTIVITY 6 · PAIRS · CALCULATION · ASSIGNMENT PATTERN

Interpolate the temperature at (0.3, 0.2) m in the solved plate

8 MIN



Given

- Solution: $T_2 = 0.482$, $T_5 = 0.386$ °C; $T_1 = T_4 = 0$ (edge).
- Mesh: a = b = 0.5 m; element 1 has nodes [1 2 5 4].

Find

- Which element contains the point; its local coordinates.
- $N_1 \ldots N_4$ at the point and their sum.
- T(0.3, 0.2).

WATCH OUT Nodal order: $\{\hat{T}^1\} = \{T_1, T_2, T_5, T_4\}$.

8 minutes. Reveal follows.

ACTIVITY 6 · SOLUTION

Element 1, local coordinates equal to global, T ≈ 0.266 °C

$$a = b = 0.5\ \text{m}, \quad (x_{\text{loc}}, y_{\text{loc}}) = (0.3,\ 0.2)\ \text{m}$$

$$N_1 = \frac{(a-x)(b-y)}{ab} = \frac{(0.2)(0.3)}{0.25} = 0.24, \qquad N_2 = \frac{x(b-y)}{ab} = \frac{(0.3)(0.3)}{0.25} = 0.36$$

$$N_3 = \frac{xy}{ab} = \frac{(0.3)(0.2)}{0.25} = 0.24, \qquad N_4 = \frac{y(a-x)}{ab} = \frac{(0.2)(0.2)}{0.25} = 0.16$$

$$\begin{aligned} T &= N_1T_1 + N_2T_2 + N_3T_5 + N_4T_4 \\ &= 0.24(0) + 0.36(0.482) + 0.24(0.386) + 0.16(0) = 0.266\ ^\circ\text{C} \end{aligned}$$

Same point shifted to (0.8, 0.2): element 2, local (0.3, 0.2), same N_i, nodal values $\{T_2, T_3, T_6, T_5\}$.



QR EXPLORER · PAIRS, ONE PHONE

Run the text’s plate on your phone

6 MIN

Scan, then do

1. Keep the 2 × 2 mesh: do T at the bottom-right corner and at the bottom middle match (2.55)?
2. Run the mesh study. From which mesh is the change below 1%? Which value would you report?
3. Make all four edges insulated. What does the Explorer say, and what does it mean physically?

Heat Conduction Explorer

`razinutm.github.io/My_Lectures/SEAA4243-FEM/labs/heat.html`

6 minutes, optional in class: if the block is running late, set it as homework. The energy-balance card shows the reaction check you will use in ANSYS (the ANSYS demo).

BLOCK 6 · 2:40 – 3:00

6

The MATLAB code and Assignment 4

Every line of the program is one of today’s steps. Then the assignment brief and the exit ticket.



BLOCK 6 · THE SCRIPT, TRACED AGAINST THE PIPELINE

Left: input and [k]. Right: assembly, Dirichlet, solve

```
% ---- input (text §2.3.1) ----
L = 1; H = 1; kx = 300; ky = 300; Q = 600;
To = 0; % Dirichlet value
Nx = 2; Ny = 2; ndof = 9;
a = L/Nx; b = H/Ny;

% ---- local conductance matrix: (2.49) ----
s1 = b*kx/(3*a) + a*ky/(3*b);
s2 = -b*kx/(3*a) + a*ky/(6*b);
s3 = -b*kx/(6*a) - a*ky/(6*b);
s4 = b*kx/(6*a) - a*ky/(3*b);
k = [s1 s2 s3 s4; s2 s1 s4 s3; s3 s4 s1 s2; s4 s3 s2 s1];
fQ = a*b*Q/4*[1; 1; 1; 1]; % (2.39)
```

```
% ---- assembly by connectivity ----
K = zeros(ndof); F = zeros(ndof,1);
e1 = [1 2 5 4]; K(e1,e1) = K(e1,e1) + k;
e2 = [2 3 6 5]; K(e2,e2) = K(e2,e2) + k;
e3 = [4 5 8 7]; K(e3,e3) = K(e3,e3) + k;
e4 = [5 6 9 8]; K(e4,e4) = K(e4,e4) + k;
F(e1) = F(e1) + fQ; % ... and e2, e3, e4

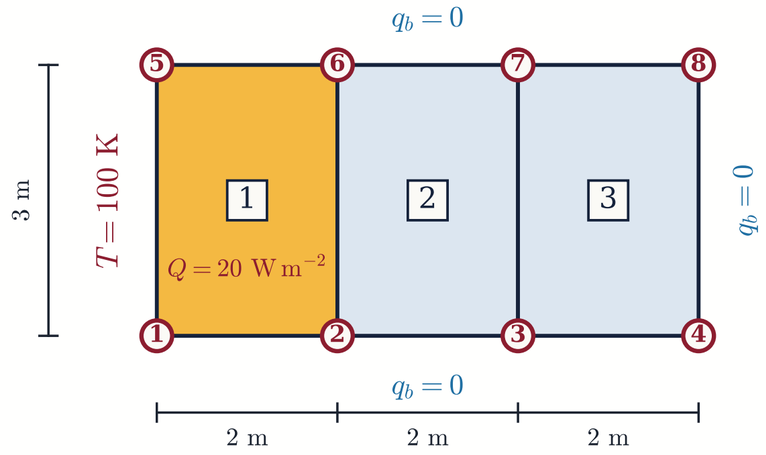
% ---- Dirichlet: row-column elimination ----
T = zeros(ndof,1);
dof_k = [1 4 7 8 9]; T(dof_k) = To;
F = F - K(:,dof_k)*T(dof_k); % known → RHS
dof_u = setdiff(1:ndof, dof_k);
T(dof_u) = K(dof_u,dof_u) \ F(dof_u); % solve
```

Run it: T(dof_u) returns 0.4821, 0.6214, 0.3857, 0.4821 — equation (2.55). Change Nx = Ny = 16 and generalise the assembly loop to reproduce the 16 × 16 column of the table.



BLOCK 6 · ASSIGNMENT 4 · HEAT TRANSFER · DUE [DATE]

Three elements, a cold left edge, one heat source



Parts

- (a) Local load vector $\{r^1\}$ for the heat source Q = 20 W m⁻² on element 1 (unit thickness).
- (b) Global [K] in terms of k, with $k_x = k_y = k$, using your Activity 3 connectivity table.
- (c) With $T_2 = T_6 = 150$ K from the solution, interpolate T at (x, y) = (1.4, 2.2) m.

$$\{r^1\} = \frac{Q\,a\,b}{4}\{1\ \ 1\ \ 1\ \ 1\}^T, \qquad T(1.4,\,2.2) = \{N(x_{\text{loc}}, y_{\text{loc}})\}\{\hat{T}^1\}^T$$

CHECK Show the connectivity table, $s_1 \ldots s_4$ with a = 2 m, b = 3 m, and the local coordinates used in (c). Marks are for the method.

ACTIVITY 7 · INDIVIDUAL · EXIT TICKET

Three questions before you leave

5 MIN

Questions

- In [k]{T̂}ᵀ = {q} + {b}, which vector does an insulated edge contribute to, and what is the contribution?
- Why must every row of [k] sum to zero, and what would a non-zero row sum tell you about your hand calculation?
- A 9-dof system has 5 Dirichlet nodes. How many equations do you solve, and what happens to the known values if they are not zero?

$$[k]\{\hat{T}\}^T = \{q\} + \{b\} = \{r\}$$

$$s_1 + s_2 + s_3 + s_4 = 0 \quad \Rightarrow \quad [k]\begin{Bmatrix}1\\1\\1\\1\end{Bmatrix} = \begin{Bmatrix}0\\0\\0\\0\end{Bmatrix}$$

Five minutes. Name and matric number; for feedback, not marks.



Same pipeline, two unknowns per node

Today in four lines

- Galerkin + IBP: $[k]\{\hat{T}\}^T = \{q\} + \{b\}$; the operator acts on the weight to its left.
- $[k]$ from four closed-form values $s_1 \ldots s_4$; rows sum to zero; $\{q\} = q_h ab/4$; edge flux splits $q_b b/2$.
- Assembly = scatter-add by connectivity; Dirichlet = move known values right, delete rows and columns.
- Post-process in local coordinates; refine the mesh before you believe a number.

Before the next session

- Assignment 4: start now; due [date].
- Read text §3.1 (plane stress PDE) — 6 pages.
- Revise from solid mechanics: plane stress vs plane strain, Hooke’s law in 2D, strain–displacement relations.
- Run the §2.3.1 script and change the mesh to 4 × 4.

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